Shashwat Siddhant

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Personal Summary

Data Scientist specialising in data analysis, **statistical modelling**, and **production ML systems**. Experienced in building detection models and **automated data platforms** across **energy** networks, **insurance** risk, **sports** and **social media** data. Strong background in **data science stack** with a focus on robust measurement, monitoring, and **decision-support analytics**.

Education

2019-20

Royal Holloway, University of London
Master of Data Science & Analytics(**1st class**)

2013-17

Manipal Institute of Technology, Manipal
Bachelor of Technology, Computer Science

Work Experience

Oct'24- Present

Data Scientist – ScottishPower Energy Networks(SPEN), Glasgow, United Kingdom

- Built and validated a hybrid **LSTM** + **Prophet demand forecasting model**, delivering ~**7.5% MAPE** across 10 primary substations and demonstrating scalable proof of value. Ongoing work.
- **Architected** and **deployed** a production-grade **Azure Synapse** (**PySpark**) pipeline that ingests network flow data via PI API, performs scalable aggregation and transformation, and publishes validated datasets to the **Open Data Portal** establishing a repeatable BAU data product.
- Developed an **analytics/reporting tool** to quantify generation loss due to curtailments from **LMS** and **ANM** in wind farms by estimating **power curve equation** using weather data and capacity ratings. **Productionized** the process as a pipeline on **Azure Synapse**.
- Analysed **Smart Meter** and **LV Monitoring** data (**SQL**, **Python**) to uncover peak demand patterns, anomalies, and weekday/weekend consumption trends across the network based on a **self-authored concept paper**.

May'21-Mar'24

Data Scientist – Vave MGA at Canopius Managing Agents Ltd, London, United Kingdom

- End-to-end automated **ETL** of **claims bordereau** data from multiple Product Partners for performance tracking using **Python** and **MS SQL Server**.
- Developed application on **Qlik Sense** for tracking **claims progression**, **portfolio analysis** and catastrophe impact using **dashboards** with intuitive visualizations and KPI's for **stakeholders**.
- Built a **catastrophe classification model** using **TF-IDF** algorithm for vectorization of text and **Random Forest** to tag **claims** based on their loss descriptions. Achieved an **F-score** of **0.91** and **accuracy** of **0.92**.
- Worked on **insurance risks data** to compute the company's **propensity** to bind it using a data set with **50+ variables** and **10M records** through **Generalized Linear Modelling**. Performed **Statistical Testing** on the variables to identify how they impact the target variable.
- Built a **Decision Trees** regressor in **Pyspark** on **Databricks** to predict **loss ratios** on **claims** using flood insurance data of **60M records** to identify pockets of potential business growth. Worked with the **Actuarial** team to implement the insights to the **pricing** engine.
- Linked policies to claims using **Fuzzy logic** technique in **Python** from claims bordereau and exposures data to track the policy portfolio across different risk locations. Achieved ~**98.5%** match.

Jun'20 – Sep'20

Data Scientist (Part-time) – In Diverse Company, London, United Kingdom

- Developed a virtual assistant for the users on the company's learning platform in the form of a **chat bot** that identifies the intent behind a user's query about the platform and gives an appropriate response. The chat bot model was built as an **Artificial Neural Network** using **Tensorflow** in **Python**.

Aug'17 – Jun'19

Data Scientist – SG Analytics, Pune, India

- Developed web crawlers built in **Python** using **Selenium** and **BeautifulSoup** to scrape social media data.
- Created dashboards for data reports using **Tableau** and **Amazon Redshift** database.
- **ETL** of social media platforms data using **Python**, **MS-Excel**, and **Amazon Redshift DB**.
- **Sentiment Analysis** of Instagram comments and tweets to understand the audience's reaction to newly launched shows using the libraries **NLTK** and **SpaCy** in Python. Achieved a **recall score** of **0.82**.
- Developed an algorithm in **Python** to identify flow of rap artists by studying the lyrics for the biggest media and entertainment firm in the US. **Communicated** effectively with **client** on regular basis.
- Built a **classification** model using **Random Forests** to analyze and predict the behavior of a wafer for **fabrication** for a firm dealing with **semiconductor devices**. Optimized the production process by bringing down the costs involved in testing phase by **24%**. Successfully **communicated with the client** and **led a team** of 3 people for this project to achieve the targets before the deadline.

Dec'16 - May'17	**Data Analyst Intern – FanZoo Sports Pvt Ltd, New Delhi, India** • Auto pick feature for fantasy gaming– **Greedy algorithm**(Knapsack) implementation in Python using sports data set. The feature was used by **61%** of the customer base. • Twitter AI Bot – Using Twitter API's and Microsoft **NLP Bot** framework to automate marketing of product which increased the **website traffic** by **250%**. • **A/B Testing** for Auto Pick Feature – Designed and executed an A/B test to compare placement of "Autopick" button on the gaming platform. Tracked engagement and win rates across 1,500+ users. • Fantasy Player Projections – Prediction model using regression analysis to predict a player's performance. Achieved an r2 value of 0.86 on the model

Technical Skills

Languages	Python, C++, C, MATLAB, R, PySpark
Data Science	Pandas, Scikit-learn, Scipy, Tensorflow, Matplotlib, Plotly, Selenium, Qlik Sense, Streamlit, Power BI, NLTK, SpaCy, MS-Excel, MS-SQL, Azure DevOps, GitHub, DBT, ArcGIS Online
Operating Systems	Android, Linux, Windows

Projects

- Worked on churn prediction model using **Logistic Regression, Random Forest** and **customer segmentation** for personalized marketing on a customer data set for a telco company under Dr. Carol Hargreaves, National University Singapore.
- **Sentiment Analysis** – Tweet fetching based on keywords and sentiment analysis of text in English, Dutch, French, Spanish, and German – as an international freelance consultant for a client in the Netherlands.
- **Credit risk analysis** of bank customers for loan approval prediction and achieved a recall score of 0.89 using **Random Forest** classifier.
- Movie Suggestion Application for Windows using C# and IMDB API as part of Code Fun Do Hackathon organized by Microsoft.
- Online ticket booking system using ASP.NET for UI and C# for backend.
- Privacy Preserving Data Mining using Python, **Apriori algorithm** and MD5.

Publications and Certifications

- Cyber security and its ten domains on Coursera by University System of Georgia.
- Co-authored a paper with fellow intern and senior Professor on '**Graphical analysis of Real-Time Data from IoT**' published in International Journal of Innovation and Scientific Research (IJISR).
- Co-authored a paper on '**Security in NFC**', published in IJTRE Volume 3, Issue 2, October 2015, ISSN(Online): 2347-4718

Extra-curriculars and Personal Skills

- Participated in *Code.Fun.Do Hackathon* organized by Microsoft on college campus. **Worked in a team** and **communicated** with all members successfully to complete the project in the 24-hour deadline.
- **Presented and explained** a self-written research paper in front of experienced PhD professors and participants during TechTatva'15, MIT, Manipal Technical Fest.
- Secured first place in Revels '16 Men's Cricket Tournament as a part of the college department team.
- Workshops organizer for Revels'15, MIT Manipal Cultural Fest. Booked halls for workshops, **negotiated, and coordinated** with the people invited to give workshops, arranged for travel tickets, and stay.
- **Finisher** in London Summer 10k race at Regent's Park in June, 2022.